Aquarium Heater



Category: **Animal Husbandry & Dairy** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Aquarium heater

What it is used for

An aquarium heater maintains water temperature within a suitable range for the aquatic species being kept. Use a correctly rated, thermostatically controlled heater and monitor temperature to prevent overheating, thermal shock, and electrical hazards.

Main parts

- Heater element and protective casing
- Thermostat or temperature control
- Temperature indicator or display, if fitted
- Power cord and plug
- Mounting clips or suction cups
- Separate aquarium thermometer

Before you start

1. Confirm the required temperature range for the species and check the heater's rated aquarium volume.
2. Inspect the heater casing, power cord, plug, and mounting accessories for cracks or damage.
3. Check that the aquarium contains enough water to cover the heater's required immersion level.
4. Position the heater according to the manufacturer's instructions, away from gravel pressure and direct contact with delicate animals.
5. Verify that the power outlet, earthing, and residual-current protection are suitable for a wet environment.
6. Allow the heater to acclimatise to water temperature as instructed before switching it on; use a separate thermometer to check current temperature.

How to use

1. Switch off and unplug the heater before installing, repositioning, or removing it.
2. Secure the heater using its mounting clips or suction cups in the recommended position.
3. Confirm that the minimum immersion line, if present, is correctly covered by water.
4. Set the thermostat to the required species-appropriate temperature.
5. Wait for any manufacturer-specified immersion or settling period before connecting power.
6. Switch on the heater and observe its indicator or display, if fitted.
7. Check aquarium temperature with a separate thermometer and compare it with the target range.
8. Monitor temperature regularly and adjust the thermostat in small increments when needed.

9. Observe aquatic animals for signs of heat stress, unusual behaviour, or distress.
10. If water temperature rises unexpectedly or the heater malfunctions, disconnect power safely and use a suitable backup temperature-control method.
11. Before water changes, maintenance, or cleaning, unplug the heater and allow it to cool for the specified period.
12. After maintenance, ensure the heater is correctly submerged and securely mounted before restoring power.

⚠ Safety

- Never operate a heater outside its specified immersion conditions.
- Never touch, remove, or expose a hot heater to air immediately after operation; allow it to cool as directed.
- Keep plugs and electrical connections dry and create a drip loop in the power cord where appropriate.
- Do not switch on a cracked heater or use one with damaged insulation or cable.
- Disconnect power before putting hands into the aquarium or servicing electrical equipment.
- Avoid sudden temperature changes that can harm aquatic animals.

Care & storage

- Remove algae and mineral deposits only after unplugging and cooling the heater, using an approved method.
- Inspect the casing, cable, thermostat, and mounting clips regularly.
- Check actual water temperature with a reliable separate thermometer.
- Replace a heater that is cracked, corroded, unreliable, or unable to maintain the set temperature.
- Store a removed heater safely and dry according to the manufacturer's instructions.

Common mistakes to avoid

- Selecting a heater with insufficient or excessive capacity for the aquarium.
- Operating the heater without the required water immersion.
- Trusting the thermostat display without checking actual water temperature.
- Handling or cleaning the heater while it is hot or connected to power.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.